Ruurd Kuiper, PhD

Assistant Professor (AI/NLP for Healthcare)

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Professional experience

Assistant Professor (2024–present)
Julius Center for Health Sciences and Primary Care (Dept. Data Science & Biostatistics), UMC Utrecht, Utrecht, The Netherlands

Senior Specialist Data Science (2023–2024)
Valcon
Applied ML/DL and NLP; agile delivery; stakeholder communication

Co-founder and Technical Lead (2022)
UPlanner (now Presurgeo)
Start-up co-founder; prototype development and validation; technical decision-making; productisation support

PhD Candidate / Researcher (2017-2022)
Utrecht University / UMC Utrecht, Utrecht, The Netherlands
Dissertation: Computer-Assisted Osteotomy Planning: Medical Imaging Techniques for Orthopaedic Surgery

Research Intern (2016-2017)
UMC Utrecht, Utrecht, The Netherlands
Topic: Automatic modelling methods for bracing of the scoliotic spine

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Education

PhD, Medical Image Analysis for Orthopaedics (2017-2022)
Utrecht University / UMC Utrecht

MSc, Biomedical Engineering (2015–2017)
Delft University of Technology

BSc, Biochemistry (Minor Finance) (2011–2014)
Delft University of Technology & Leiden University

Research interests

- Clinical NLP and AI for healthcare: development, evaluation, implementation
- Generative/foundation models (LLMs and multimodal models) for clinical tasks
- Trustworthy AI: validation protocols, reliability and failure analysis
- Medical imaging and computer-assisted planning for orthopaedics
- Alternative language-model architectures and generative paradigms

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Grants and funding

- Applied Data Science (Utrecht University): Exploring Diffusion Models for Enhanced Medical Text Summarization (main applicant; €5k)
- Kansen voor West: co-authored grant application (~€600k) supporting translation of PhD research into a start-up context
- RvO start-up grant: co-authored grant application (~€40k); beneficiary via salary support at UPlanner

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Teaching experience

Co-teacher, Machine Learning: Applications in Medicine - 2026
Taught one day of the five-day course

Co-teacher, Machine Learning in Healthcare - 2025
Taught one day of the five-day course

Instructor / course developer, Large Language Models in Healthcare - 2025
SSACAB - University of the Witwatersrand (Johannesburg, South Africa)
Five-day short course (20–24 October 2025) with practicals

Teaching assistant, Advanced Machine Learning Methods in Healthcare - 2025
University of the Witwatersrand (Johannesburg, South Africa)

Co-teacher, Big Data course / LLM practical - 2024, 2025
Short course, UMC Utrecht

Teaching assistant, Missing Data course - 2024
MSc Epidemiology, UMC Utrecht

Teaching practicals, Medical Imaging Techniques - 2018 - 2021
MSc Image Analysis, UMC Utrecht

Supervision and mentoring

- Supervised eight MSc thesis students in healthcare AI, medical NLP, and language-model development

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Publications (selected)

Doctoral thesis

- **Kuiper, R.** (2023). Computer-Assisted Osteotomy Planning: Medical Imaging Techniques for Orthopaedic Surgery. Utrecht University.

First author journal articles

- **Kuiper, R. J. A.**†, & Leeuwenberg, A. M.† (2026). Natural language processing and language models for Dutch clinical text: a systematic review. *Zenodo*. (Submitted; preprint; †shared first authors; accepted for CLIN36).
- **Kuiper, R. J. A.**†, de Hond, A. A. H.†, Spiero, I., Chen, Y.-W., Idema, D. L., Damen, J. A. A., van Smeden, M., Moons, K. G. M., & Leeuwenberg, A. M. (2026). Validation of Large Language Models in Healthcare: a Scoping Review. *Zenodo*. (Preprint; †shared first authors).
- **Kuiper, R. J. A.**, Wolters, E. J., de Vries, S., Erler, N. S., van Smeden, M., Kant, I. M. J., & Marsman, M. M. (2026). Development and validation of AI-assisted generation of anesthesiology preoperative consultation summaries. *Zenodo*. (Preprint).
- **Kuiper, R. J. A.**†, Bartels, R.†, et al. (2026). Large Language Models in Healthcare: on Realistic Expectations and Quality Control. *Heliyon*. (Submitted; †shared first authors).
- **Kuiper, R. J. A.**, Seevinck, P. R., Viergever, M. A., Weinans, H., & Sakkers, R. J. B. (2023). Automatic Assessment of Lower-Limb Alignment from Computed Tomography. *The Journal of Bone and Joint Surgery. American Volume*, 105(9), 700–712.
- **Kuiper, R. J. A.**, Colaris, J. W., Stockmans, F., van Es, E. M., Viergever, M. A., Seevinck, P. R., Weinans, H., & Sakkers, R. J. B. (2023). Impact of bone and cartilage segmentation from CT and MRI on both bone forearm osteotomy planning. *International Journal of Computer Assisted Radiology and Surgery*, 18(12), 2307–2318.
- **Kuiper, R. J. A.**, Sakkers, R. J. B., van Stralen, M., Arbabi, V., Viergever, M. A., Weinans, H., & Seevinck, P. R. (2022). Efficient cascaded V-net optimization for lower extremity CT segmentation validated using bone morphology assessment. *Journal of Orthopaedic Research*, 40(12), 2894–2907.
- **Kuiper, R. J. A.**, van Stralen, M., Sakkers, R. J. B., Bergmans, R. H. J., Zijlstra, F., Viergever, M. A., Weinans, H., & Seevinck, P. R. (2021). CT to MR registration of complex deformations in the knee joint through dual quaternion interpolation of rigid transforms. *Physics in Medicine and Biology*, 66.

- **Kuiper, R. J. A.** (2017). A Finite Element Approach to Brace Treatment for the Scoliotic Spine. *Unpublished internship research*, UMC Utrecht.

Full conference papers

- **Kuiper, R. J. A.**, de Groot, L., van Es, B., van Smeden, M., & Bagheri, A. (2025). LAD: LoRA-Adapted Diffusion. *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP): System Demonstrations*, 97–110.
- de Groot, L., **Kuiper, R. J. A.**, & Bagheri, A. (2025). A Survey of Discrete Diffusion for Text and Genomic Sequence Generation. *Proceedings of the 37th Benelux Conference on Artificial Intelligence (BNAIC) and the 34th Belgian-Dutch Conference on Machine Learning (BeNeLearn)*.

Co-authored publications and conference abstracts

- Nguyen, H. C., Bax, E., Custers, R. J. H., van Egmond, N., **Kuiper, R. J. A.**, … (2026). Progression of bone and joint space deformity in patients with mild knee osteoarthritis: Data from the IMI-APPROACH cohort. *Osteoarthritis and Cartilage Open*, 8(2), 100762.
- de Hond, A. A. H., **Kuiper, R. J. A.**, Spiero, I., Chen, Y.-W., Idema, D. L., Damen, J. A. A., van Smeden, M., Moons, K. G. M., & Leeuwenberg, A. M. (2026). A scoping review on validation methods and frameworks for Large Language Models in healthcare. *Diagnostic and Prognostic Research*, 10(Suppl 1), abstract P74 (MEMTAB 2025).
- **Kuiper, R. J. A.**, Carriero, A., & van Smeden, M. (2026). Visualizing SHAP values associated with hallucinations in language models. *Diagnostic and Prognostic Research*, 10(Suppl 1), abstract P26 (MEMTAB 2025).
- Veerman, Q. W. T., Tuijthof, G. J. M., Verdonschot, N., … **Kuiper, R. J. A.**, … (2025). A structured framework for standardized 3D leg alignment analysis: an international Delphi consensus study. *Knee Surgery, Sports Traumatology, Arthroscopy*, 33(6), 2276–2292.
- Bax, E. A., Kerkhof, J. A. J., van Egmond, N., **Kuiper, R. J. A.**, … (2025). The Impact of varus and valgus alignment on knee cartilage quality assessed by magnetic resonance imaging. *The Knee*, 57, 477–487.
- Nguyen, C., Bax, E., Custers, R., van Egmond, N., **Kuiper, R.**, Arbabi, V., … (2025). Progression of bone and intra-articular joint space deformity in patients with mild knee osteoarthritis: data from the IMI-APPROACH cohort. *Osteoarthritis and Cartilage*, 33, S363.
- Bax, E., Custers, R. J., van Egmond, N., Rayegan, H., **Kuiper, R.**, … (2025). Bony Varus Malalignment Accelerates Knee Osteoarthritis Progression: Insights from Joint Line Convergence Angle in the IMI-APPROACH Cohort. *Osteoarthritis and Cartilage*, 33, S190.
- Willemsen, K., Ketel, M. H. M., Zijlstra, F., Florkow, M. C., **Kuiper, R. J. A.**, … (2021). 3D-printed saw guides for lower arm osteotomy, a comparison between a synthetic CT and CT-based workflow. *3D Printing in Medicine*, 7, 1–12.

- Anadria, D., Dobbe, R., Giachanou, A., **Kuiper, R.**, Bartels, R., van Amsterdam, W., de Troya, I. M., Zürcher, C., & Oberski, D. (2025). From Data-Driven to Purpose-Driven Artificial Intelligence: Systems Thinking for Data-Analytic Automation of Patient Care. *arXiv preprint arXiv:2506.13584*.
- **Kuiper, R. J. A.**, Willemsen, K., Weinans, H., Viergever, M. A., Seevinck, P. R., & Sakkers, R. J. B. (2021). Semi-Automatic Osteotomy Planning with Optimal Joint Alignment. *39th Annual Meeting of the European Paediatric Orthopaedic Society*, abstract EP-097.
- **Kuiper, R. J. A.**, van Stralen, M., Zijlstra, F., Weinans, H., Roth, K. C., Colaris, J. W., Sakkers, R. J. B., & Seevinck, P. R. (2019). A Semi-Automatic Workflow for CT-MRI Registration of Complex Deformations Induced by Interscan Radio-Ulnar Rotations. *7th Dutch Bio-Medical Engineering Conference*, Egmond aan Zee, the Netherlands.

Academic service

- Coordinator, NLP@U: collaboration among NLP researchers at Utrecht University and UMC Utrecht
- EWUU alliance: contributing UMC Utrecht clinical-AI expertise to collaboration with Eindhoven, Wageningen, and Utrecht universities
- Co-organiser / Program Chair, Alternative Architectures and Paradigms (AAP) workshop, INLG 2026, with Robert Bagheri

Peer reviewer (selected journals)

- PLOS Digital Health
- BioMed Research International
- Artificial Intelligence in Medicine
- International Journal of Computer Assisted Radiology and Surgery (IJCARS)
- BMC Musculoskeletal Disorders
- Journal of Orthopaedic Surgery and Research
- Scientific Reports (Nature Portfolio)
- Biomedical Engineering / Biomedizinische Technik

Certifications

University Teaching Qualification (BKO), ongoing as of September 2026

eBROK® Basiscursus + Verdieping nWMO, NFU – Nederlandse Federatie van Universitair Medische Centra (Issued Jan 2026; expires Jan 2028)

Natural Language Processing Specialization, Coursera (Issued Sep 2023)

SQL Essential Training, LinkedIn Learning (Issued May 2023)

Skills

Programming: Python, MATLAB, R, Swift, C#

ML/DL: PyTorch, TensorFlow; computer vision; NLP; evaluation and validation pipelines

Data/engineering: NumPy/SciPy, scikit-learn, pandas; software development and prototyping; agile/Scrum

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